

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangam, Machhe, Belagavi-590018



Scheme of Teaching and Examinations 2025

Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS)
(Effective from the academic year 2025-26)

B.E. in Industrial IoT,
Scheme of Teaching and Examinations 2025
Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)

III SEMESTER															
Sl. No	Course	Course Code	Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)		Teaching & Learning Scheme				Examination				Credits	
						CI		LI	TW & SL	# TH/S	Duration in hours	CIE Marks	SEE Marks		Total Marks
						L	T	P	SL						
1	ASC/PCC	1BIO301	Transforms, Differential Equations & Linear Algebra	TD /PSB: Mathematics / Specific department		42	28	0	50	120	3	50	50	100	4
2	IPCC	1BIO302	Data Structures & Algorithms	TD/PSB		42	0	28	50	120	3	50	50	100	4
3	PCC	1BIO303	Digital Electronics and Logic Design	TD/PSB		42	0	28	50	120	3	50	50	100	4
4	PCC	1BIO304	Sensors & Transducers	TD/PSB		42	0	0	48	90	3	50	50	100	3
5	PCC	1BIO305	Fundamentals of Industrial IoT	TD/PSB		42	0	0	48	90	3	50	50	100	3
6	PCCL	1BIOL306	Sensors & IoT Devices Lab	TD/PSB		0	0	28	2	30	2	50	50	100	1
7	AEC	1BIOL307x	Ability Enhancement Course Laboratory**	TD/PSB		0	0	28	2	30	2	50	50	100	1
8	SDC	1BCP308	Community Project (Project-Based Learning) / Societal Project	Any Department/ Respective Engineering Dept.		0	0	0	30	30	2	50	50	100	1
9	NCMC	1BNSS309	National Service Scheme (NSS)	Campus	NSS coordinator	0	0	0	28	28	--	100	---	100	PP
		1BPE309	Physical Education (PE) (Sports and Athletics)												
		1BYOG309	Yoga												
		1BMUK309	Music												
Total												500	400	900	21
10	NCMC	1BMATDIP310	Mathematics course for Lateral Entry Students	TD/PSB -Maths Dept		14	0	0	14	28	3	100	---	100	PP
Total										686		600	400	1000	21

S- Total Hours per Semester, TW & SL- Term Work & Self Learning

Ability Enhancement Course (Laboratory) 1BxxL307x			
1BIOL307A	Python Programming for IoT	1BIOL307C	Fundamentals of Linux & Shell Scripting (IoT-Focused)
1BIOL307B	PCB Design & Soldering Skills	1BIOL307D	MATLAB / Simulink for IoT
** The course 1BXXL307 – Ability Enhancement Course Laboratory can be offered either as a single compulsory course. Alternatively, the course 1BXXL307 – Ability Enhancement Course Laboratory shall be offered as multiple elective options under the course codes 1BXXL307x (where $x = A, B, C, D$). Note to Chairpersons: If only one course is selected, the title of the course may please be entered at serial number 7 of the Scheme of Teaching and Examinations and the above table along with this row shall be deleted. In case, multiple courses are selected the above table shall be filled with the course titles and this row shall be deleted.			

Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS) (Effective from the academic year 2025-26)

	IV SEMESTER																	
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching & Learning Scheme				Examination				Credits					
					CI		LI	# TW & SL	TH/S	Duration in hours	CIE Marks	SEE Marks		Total Marks				
															L	T	P	
1	ASC/PCC	1BIO401	Numerical Methods, Probability & Statistics)	TD/PSB:	42	0	0	48	90	3	50	50	100	3				
2	IPCC	1BIO402	Industrial Automation & PLC	TD/PSB	42	0	28	50	120	3	50	50	100	4				
3	PCC	1BIO403	Database Management Systems	TD/PSB	42	0	28	50	120	3	50	50	100	4				
4	PCC	1BIO404	Microcontrollers & Embedded Systems	TD/PSB	42	0	0	48	90	3	50	50	100	3				
5	PCCL	1BIOL405	Embedded Systems Lab	TD/PSB	0	0	28	02	30	2	50	50	100	1				
6	AEC	1BIOL406	Ability Enhancement Course Laboratory**	TD/PSB	0	0	28	02	30	2	50	50	100	1				
7	BSC	1BIO407	Biology for Engineers	TD / PSB	28	0	0	32	60	3	50	50	100	2				
8	SDC	1BEP408	Environmental Science Project	TD/PSB	0	0	0	30	30	3	50	50	100	1				
9	PCC/PEC	1BIO409	Signals & Systems		42	0	0	48	90	3	50	50	100	3				
10	NCMC	1BNSK409	National Service Scheme (NSS)	Campus	NSS coordinator	Physical Education Director	Yoga Teacher	Music Teacher	0	0	0	28	28	--	100	---	100	PP
		1BPEK409	Physical Education (PE) (Sports and Athletics)															
		1BYOK409	Yoga															
		1BMUS409	Music															
Total									688		550	450	1000	22				
11	NCMC	1BMATDIP410	Mathematics course for Lateral Entry Students	TD/PSB -Maths Dept	14	0	0	14	28	--	100	--	100	PP				
Total									716		650	450	1100	22				
	Ability Enhancement Course (Laboratory) 1BxxL406x																	
1BIOL406A		IoT Device Programming with Arduino & ESP32			1BIOL406C		Dashboard & Data Visualization Lab											

1BIOL406B	Signal processing using Matlab	1BIOL406D	Industrial Safety & IoT Standards
	# TH/S- Total Hours per Semester, TW & SL- Term Work & Self Learning, ** The course 1BXXL406x – Ability Enhancement Course Laboratory can be offered either as a single compulsory course. Alternatively, the Ability Enhancement Course Laboratory course can be offered with multiple elective options under the course codes 1BXXL406x (where $x = A, B, C, D$).		